

System Security Plan — VSG-100 Secure Network Gateway (NIST SP 800-171)

Technical writing sample by Nathan B. Smith. This document contains no export-controlled, CUI, or classified information. The equipment described is representative of real systems but is entirely fictional.

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SAMPLE

Disclaimer

This document set is a writing sample and does not describe a real system, company, or security configuration.

This System Security Plan and its supporting topics were created by Nathan B. Smith solely to demonstrate technical writing skill in authoring compliance-mapped security documentation. They are not associated with any actual government or commercial program.

- **Fictional company:** TechCoaches Information Systems Design is a fictional firm created for this and related writing samples. It is not a real business entity.
- **Fictional equipment:** The VSG-100 Secure Network Gateway is a fictional system. It does not represent any real product, platform, or vendor architecture.
- **No CUI:** This document set contains no actual Controlled Unclassified Information. All system details, control implementation statements, and diagrams are illustrative placeholders.
- **No export-controlled content:** This document set contains no technical data subject to the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR). No portion of this sample is intended to convey, and none should be interpreted as conveying, export-controlled technical information.

Content in this document set should be evaluated only as a demonstration of DITA authoring structure, NIST SP 800-171 documentation conventions, and technical writing practice.

About this system security plan

The System Security Plan (SSP) describes how the VSG-100 Secure Network Gateway implements the security requirements of NIST SP 800-171 to protect Controlled Unclassified Information (CUI).

A System Security Plan is the foundational document that describes the security requirements in place, or planned, for an information system, and the controls in place, or planned, to meet those requirements. For contractors handling CUI on behalf of the Department of Defense, an SSP is required under DFARS clause 252.204-7012 and is the reference document used during a Cybersecurity Maturity Model Certification (CMMC) assessment.

This SSP covers the VSG-100 Secure Network Gateway™, a fictional tactical network appliance developed for this documentation sample. The SSP describes:

- The system boundary and the CUI that flows through it
- How each applicable NIST SP 800-171 control family is implemented
- Relationships with external systems and service providers
- Roles and responsibilities for maintaining the system's security posture



Note: This document is a portfolio writing sample. See [Disclaimer](#) on page 3 for the full disclaimer covering the fictional company, fictional equipment, and the absence of CUI or export-controlled content in this set.

System description and boundary

Identifying information, purpose, and system boundary for the VSG-100 Secure Network Gateway.

Field	Description
System Name	VSG-100 Secure Network Gateway (fictional)
System Owner	TechCoaches Information Systems Design
System Purpose	Provides encrypted gateway services between a tactical network segment and a program office network,

Field	Description
	enforcing boundary and access controls for CUI in transit.
Information Type	Controlled Unclassified Information (CUI) — Export Controlled and Controlled Technical Information categories
System Boundary	Includes the VSG-100 appliance, its management console, and the two network interfaces carrying CUI traffic. Excludes the tactical network segment and program office network themselves, which are separately authorized.
External Connections	One inbound connection from the tactical network segment; one outbound connection to the program office network, via the encrypted interfaces configured in Configure boundary protection on the VSG-100 on page 6.
Authorizing Official	[Program-designated Authorizing Official]

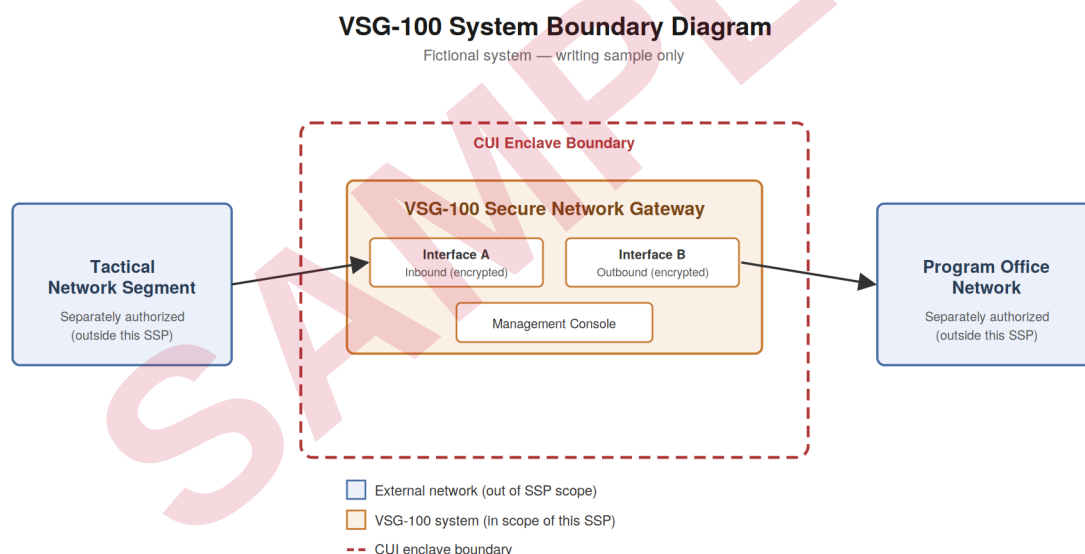


Figure 1: VSG-100 system boundary diagram



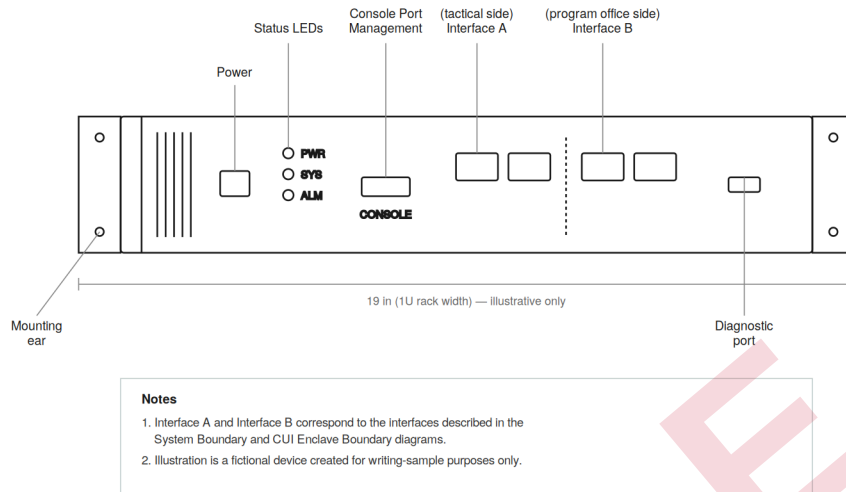
Note: All values in this table are fictional. See [Disclaimer](#) on page 3 for the full disclaimer.

VSG-100 front panel

Physical layout and connector identification for the VSG-100 front panel.

VSG-100 Secure Network Gateway — Front Panel

Fictional device — writing sample only — line drawing not to scale

**Figure 2: VSG-100 front panel**

Item	Description
Power	Power connector. Confirm the unit is powered down before connecting or disconnecting.
Status LEDs	PWR, SYS, and ALM indicators. Refer to the VSG-100 maintenance manual for indicator states and troubleshooting guidance.
Management console port	Administrator access point used to reach the VSG-100 management console referenced throughout this SSP.
Interface A	Tactical-side network interface. Corresponds to Interface A in System description and boundary on page 3 and Configure boundary protection on the VSG-100 on page 6.
Interface B	Program-office-side network interface. Corresponds to Interface B in System description and boundary on page 3 and Configure boundary protection on the VSG-100 on page 6.
Diagnostic port	Local diagnostic access only. This port sits outside the CUI enclave boundary; do not use it to pass CUI traffic.
Mounting ears	Standard 19-inch, 1U rack-mount hardware.



Note: This illustration is fictional and was created to demonstrate equipment line-drawing conventions. See [Disclaimer](#) on page 3 for the full disclaimer.

NIST SP 800-171 control families addressed in this plan

NIST SP 800-171 organizes its 110 security requirements into 14 control families. This SSP addresses a representative subset of those families for the VSG-100 system.

Each control family groups related security requirements. A complete SSP addresses all 14 families; this sample focuses on three families that are most relevant to a network gateway appliance handling CUI in transit and at rest.

Access Control (AC)

Requirements governing who and what can access the system, and under what conditions. Covered in [Configure access control on the VSG-100](#) on page 6.

System and Communications Protection (SC)

Requirements for protecting information at the system's external and internal boundaries, including encryption of CUI in transit. Covered in [Configure boundary protection on the VSG-100](#) on page 6.

Configuration Management (CM)

Requirements for establishing and maintaining baseline configurations and controlling changes to the system. Implementation status is summarized in [Control implementation status](#) on page 7.

Every control in an SSP is addressed in one of two ways: it is fully implemented, or its gap and remediation timeline are documented in a Plan of Action and Milestones (POA&M). This sample assumes full implementation for illustration.

Configure access control on the VSG-100

Configure role-based access control and session enforcement on the VSG-100 to satisfy NIST SP 800-171 Access Control (AC) requirements.

- Administrator-level access to the VSG-100 management console
- An approved list of authorized users and their assigned roles

This procedure implements AC.L2-3.1.1 (limit system access to authorized users), AC.L2-3.1.2 (limit access to the types of transactions authorized users are permitted to execute), and AC.L2-3.1.11 (terminate a user session after a defined condition).

1. Select **Access Control** in the VSG-100 management console.
2. Create a role for each authorized function (Administrator, Operator, Auditor) and assign only the permissions required for that function.
Don't assign Administrator privileges to accounts used for routine monitoring or reporting.
3. Assign each authorized user to the role that matches their approved function on the authorized user list.
4. Set the **Session Timeout** value to the organization's approved idle-session limit.
An idle session that exceeds this value is automatically terminated, satisfying AC.L2-3.1.11.
5. Select **Login Attempt Lockout**, and set the maximum number of consecutive failed login attempts allowed before an account is locked.
6. Save the configuration and review the resulting role assignments in the **Access Summary** report.

Access to the VSG-100 is restricted to authorized users, each limited to the transactions their assigned role permits, with idle sessions terminated automatically.

Configure boundary protection on the VSG-100

Configure the VSG-100's external boundary and encryption settings to satisfy NIST SP 800-171 System and Communications Protection (SC) requirements.

- Administrator-level access to the VSG-100 management console
- An approved network diagram identifying the CUI enclave boundary

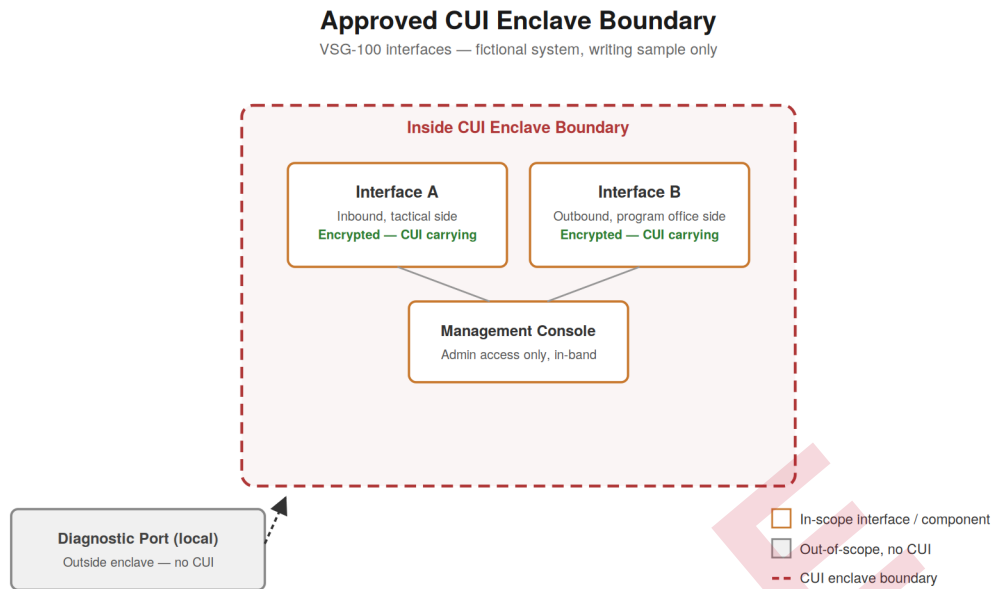


Figure 3: Approved CUI enclave boundary

This procedure implements SC.L2-3.13.1 (monitor and control communications at external boundaries) and SC.L2-3.13.8 (implement cryptographic mechanisms to prevent unauthorized disclosure of CUI during transmission).

1. Select **Boundary Protection** in the VSG-100 management console.
2. Define the CUI enclave boundary by identifying which network interfaces carry CUI traffic, using the approved network diagram as reference.
3. Turn on the managed interface filter, apply a deny-by-default rule set, and then add explicit allow rules only for approved traffic flows.
4. Under **Transport Encryption**, select an approved FIPS-validated cryptographic module for all CUI-carrying interfaces.
Interfaces without an approved cryptographic module selected will not pass CUI traffic once enforcement is enabled.
5. Select **Enforce Encryption** for all interfaces identified as part of the CUI enclave boundary.
6. Save the configuration and confirm the **Boundary Status** report shows all CUI interfaces as filtered and encrypted.

Communications at the VSG-100's external boundary are monitored and controlled, and CUI is protected by an approved cryptographic mechanism during transmission.

Control implementation status

Implementation status for the NIST SP 800-171 controls addressed in this SSP sample.

Control ID	Requirement	Status	Implementation Reference
AC.L2-3.1.1	Limit system access to authorized users	Implemented	Configure access control on the VSG-100 on page 6

Control ID	Requirement	Status	Implementation Reference
AC.L2-3.1.2	Limit access to authorized transaction and function types	Implemented	Configure access control on the VSG-100 on page 6
AC.L2-3.1.11	Terminate a user session after a defined condition	Implemented	Configure access control on the VSG-100 on page 6
SC.L2-3.13.1	Monitor and control communications at external boundaries	Implemented	Configure boundary protection on the VSG-100 on page 6
SC.L2-3.13.8	Implement cryptographic mechanisms to prevent unauthorized disclosure of CUI during transmission	Implemented	Configure boundary protection on the VSG-100 on page 6
CM.L2-3.4.1	Establish and maintain baseline configurations	Planned — POA&M item CM-01, target Q1	Not yet documented in this sample



Note: The final row illustrates how an SSP documents an outstanding control via a Plan of Action and Milestones (POA&M) entry rather than claiming full implementation.